MAX-PLANCK-GESELLSCHAFT

Lyapunov spectrum of nonautonomous stochastic systems driven by a fractional Brownian noise

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Contents

References

- joint work with Nguyen Dinh Cong (IMH-VAST) and Phan Thanh Hong (TLU)
- NDC, LHD, PTH: Nonautonomous Young differential equations revisited. Preprint 2017
- NDC, LHD, PTH: Lyapunov spectrum for SDE driven by fBm. Preprint 2017

p-var norms, Hölder norms

Let $0 \leq T_1 < T_2 < \infty$, $C^{r-\mathrm{var}}([T_1, T_2], \mathbb{R}^d)$ be the Banach space of bounded $r-$ variation continuous functions on $[T_1, T_2]$ having values in $\mathbb{R}^d$ with finite norm

$$\|u\|_{r-\mathrm{var},[\mathrm{T}_1,\mathrm{T}_2]} = |u_{T_1}| + \|\!|u|\!\|_{r-\mathrm{var},[\mathrm{T}_1,\mathrm{T}_2]} < \infty,$$

where

$$\|\!|u|\!\|_{p-\mathrm{var},[\mathrm{T}_1,\mathrm{T}_2]} = \left(\sup_{\Pi(T_1,T_2)} \sum_{i=0}^{n-1} |u_{t_{i+1}} - u_{t_i}|^r \right)^{1/r}$$

For each $0 < \alpha < 1$, we denote by $C^{\alpha-\mathrm{Hol}}([T_1, T_2], \mathbb{R}^d)$ the space of Holder continuous functions on $[T_1, T_2]$ equipped with the norm

$$\|u\|_{\alpha-\mathrm{Hol},[\mathrm{T}_1,\mathrm{T}_2]} := |u_{T_1}| + \sup_{T_1 \leq s < t \leq T_2} \frac{|u(t) - u(s)|}{(t-s)^\alpha}.$$

We also denote by $\|\cdot\|_{\infty,[T_1,T_2]}$ the supnorm of continuous functions on $[T_1, T_2]$. It is obvious that $\forall u \in C^{\alpha-\mathrm{Hol}}([T_1, T_2], \mathbb{R}^d)$ with $\alpha = 1/r$.

$$\|\!|u|\!\|_{r-\mathrm{var},[\mathrm{T}_1,\mathrm{T}_2]} \leq (T_2 - T_1)^\alpha \|\!|u|\!\|_{\alpha-\mathrm{Hol},[\mathrm{T}_1,\mathrm{T}_2]}$$

Young integrals

Now, consider $x \in C^{q-\mathrm{var}}([T_1, T_2], \mathbb{R}^{d\times m})$ and $\omega \in C^{p-\mathrm{var}}([T_1, T_2], \mathbb{R}^m)$, $p, q \geq 1$, if Riemann-Stieltjes sums for finite partition $\Pi = \{T_1 = t_0 < t_1 < \cdots < t_n = T_2\}$ of $[T_1, T_2]$ and any $\xi_i \in [t_i, t_{i+1}]$

$$S_\Pi \quad := \quad \sum_{i=0}^{n-1} x(\xi_i)(\omega(t_{i+1}) - \omega(t_i)),$$

converges as the mesh $|\Pi| := \max\limits_{0\leq i\leq n-1} |t_{i+1} - t_i|$ tends to zero, we call the limit is the *Young integral* of x w.r.t ω on $[T_1, T_2]$ denoted by $\int_{T_1}^{T_2} x(t)d\omega(t)$. It is well known that if $p, q \geq 1$ and $\frac{1}{p} + \frac{1}{q} > 1$, the Young integral $\int_a^b x_t d\omega_t$ exists, and satisfies the so-called Young-Loeve estimate

$$\left| \int_s^t x(u)d\omega(u) - x(s)[\omega(t) - \omega(s)] \right| \leq K \left\| x \right\|_{q-\mathrm{var},[\mathrm{s,t}]} \left\| \omega \right\|_{p-\mathrm{var},[\mathrm{s,t}]}, \tag{1.1}$$

where

$$K := (1 - 2^{1-\theta})^{-1}, \qquad \theta := \frac{1}{p} + \frac{1}{q} > 1. \tag{1.2}$$

SDE driven by fBm

Consider the equation

$$x(t) = x_{t_0} + \int_{t_0}^{t} A(s)x(s)ds + \int_{t_0}^{t} C(s)x(s)d\omega(s), \tag{2.1}$$

in which $A \in C([t_0, t_0+T], \mathbb{R}^{d\times d})$, $C \in C^{q-\text{var}}([t_0, t_0+T], \mathbb{R}^{d\times d})$, $\omega \in C^{p-\text{var}}([t_0, t_0+T], \mathbb{R})$ with some $1 < p < 2$, $p < q$ and $\frac{1}{q} + \frac{1}{p} > 1$. We will show that (2.1) has a unique solution in $C^{q-\text{var}}([t_0, t_0+T], \mathbb{R}^d)$. We now consider $x \in C^{q-\text{var}}([a,b], \mathbb{R}^d)$ with some $[a,b] \subset [t_0, t_0+T]$. Define the mapping given by

$$\begin{aligned} F(x)(t) &= x(a) + I(x)(t) + J(x)(t) \qquad (2.2)\\ &:= x(a) + \int_a^t f(s,x(s))ds + \int_a^t g(s,x(s))d\omega(s), \quad \forall t \in [a,b]. \end{aligned}$$

Lemma

Let $C \in C^{q-\mathrm{var}}([t_0, t_0+T], \mathbb{R}^{d\times d})$, $x \in C^{q-\mathrm{var}}([t_0, t_0+T], \mathbb{R}^d)$, *then for all* $s < t$ *in* $[t_0, t_0+T]$

$$\left\|Cx\right\|_{q-\mathrm{var},[s,t]} \leq \|C\|_{\infty,[s,t]} \left\|x\right\|_{q-\mathrm{var},[s,t]} + \|x\|_{\infty,[s,t]} \left\|C\right\|_{q-\mathrm{var},[s,t]}$$

Proposition

Assuming that $x \in C^{q-\mathrm{var}}([t_0, t_0+T], \mathbb{R}^d)$. *Then* $F(x) \in C^{p-\mathrm{var}}([t_0, t_0+T], \mathbb{R}^d)$ *and the following estimates hold for every* $[s,t] \subset [t_0, t_0+T]$

$$(i) \quad \|Fx\|_{p-\mathrm{var},[s,t]} \leq |x(s)| + \Big[\|A\|_{\infty,[s,t]}(t-s) + (K+1)\|C\|_{q-\mathrm{var},[s,t]} \left\|\omega\right\|_{p-\mathrm{var},[s,t]}\Big] \|x\|$$

$$(ii) \quad \|Fx - Fy\|_{p-\mathrm{var},[s,t]} \leq |x(s) - y(s)| + \Big[\|A\|_{\infty,[s,t]}(t-s) + (K+1)\|C\|_{q-\mathrm{var},[s,t]} \left\|\omega\right\|_{p-\mathrm{var},[s,t]}\Big] \|x-y\|_{q-\mathrm{var}}$$

where K *is defined in* (1.2).

Theorem

Fix $[t_0, t_0+T]$ and assuming that $A \in C([t_0,t_0+T],\mathbb{R}^{d\times d})$, $C \in C^{q-\mathrm{var}}([t_0,t_0+T],\mathbb{R}^{d\times d})$ with $p<q$ and $\frac{1}{q}+\frac{1}{p}>1$. Then equation (2.1) has a unique solution $x(\cdot,t_0,\omega,x_{t_0})$ in the space $C^{q-\mathrm{var}}([t_0,t_0+T],\mathbb{R}^d)$. Moreover, the solution belongs to the space $C^{0,q-\mathrm{var}}([t_0,t_0+T],\mathbb{R}^d)$.

Corollary

Under the assumption of Theorem 2, the solution $x(.,t_0,\omega,x_0)$ of (2.1) satisfies with $\eta = -\log(1-\mu)$, C_1, C_2 are constants depends on μ, M^*

$$(i)\quad \|x(.,t_0,\omega,x_{t_0})\|_{\infty,[t_0,t_0+T]} \leq |x_{t_0}| e^{\eta[2+(\frac{2M^*}{\mu})^p (T^p + \|\!|\omega|\!\|^p_{p-\mathrm{var},[t_0,t_0+T]})]}, \tag{2.3}$$

$$(ii)\quad \|x(.,t_0,\omega,x_{t_0})\|_{q-\mathrm{var},[t_0,t_0+T]} \leq |x_{t_0}| \left(1 + e^{C_2(\mu,M^*)(T^p+\|\!|\omega|\!\|^p_{p-\mathrm{var},[t_0,t_0+T]})}\right), \tag{2.4}$$

Now, we consider ω varying as an element of $C^{p-\mathrm{var}}([t_0, t_0+T])$ with $p-$var norm and defined the solution mapping

$$\begin{aligned} X : [t_0, t_0+T] \times \mathbb{R}^d \times C^{p-\mathrm{var}}([t_0, t_0+T], \mathbb{R}) &\longrightarrow C^{q-\mathrm{var}}([t_0, t_0+T], \mathbb{R}^d) \\ (a, x_0, \omega) &\mapsto X(\cdot, a, x_0, \omega). \end{aligned}$$

Proposition

The solution map is continuous w.r.t (a, x_0, ω).

If $\omega \in C^{1/p-\mathrm{Hol}}([t_0, t_0+T], \mathbb{R}) \subset C^{p-\mathrm{var}}([t_0, t_0+T], \mathbb{R}^d)$ then it is easy to see that the solution is $1/q-$ Holder continuous and

$$X : [t_0, t_0+T] \times \mathbb{R}^d \times C^{1/p-\mathrm{Hol}}([t_0, t_0+T], \mathbb{R}) \longrightarrow C^{1/q-\mathrm{Hol}}([t_0, t_0+T], \mathbb{R}^d)$$

is continuous.

Generation of stochastic two parameter flow

We construct a similar the metric dynamical system $(\Omega, \mathcal{F}, \mathbb{P}, (\theta_t)_{t\in\mathbb{R}})$, where Ω can be chosen as $C^0(\mathbb{R},\mathbb{R})$, equipped with the compact open topology in $\mathbb{R}$, $\mathcal{F}$ as $\mathcal{B}(C^0)$, the associated Borel-σ-algebra and $\mathbb{P}$ as the Wiener measure. θ can be constructed as the Wiener shift

$$\theta_t m(\cdot) = m(t+\cdot) - m(t), \forall t \in \mathbb{R}, m \in \Omega.$$

For $1/2 < \nu < H, p := \frac{1}{\nu}$, we then consider $p-$var version for the construction of fBm:

$$B^H : (\Omega, \mathcal{F}, \mathbb{P}, (\theta_t)_{t\in\mathbb{R}}) \longrightarrow (C^{0,p-\mathrm{var}}(\mathbb{R},\mathbb{R}), \mathcal{G})$$

in which $C^{0,p-\mathrm{var}}(\mathbb{R},\mathbb{R})$ is the space of real functions belongs to $C^{0,p-\mathrm{var}}([a,b],\mathbb{R})$ on every closed interval $[a,b]$. By embedding we may define $B^H(\omega) = \omega$, for all $\omega \in (\Omega^* := C^{0,p-\mathrm{var}}(\mathbb{R},\mathbb{R}), \mathcal{G}, \mathbb{P}_H, (\theta_t)_{t\in\mathbb{R}})$ with probability measure $\mathbb{P}_H$
Since $\omega \mapsto \|\omega\|_{p-\mathrm{var},[\mathrm{a,b}]}$ and $\omega \mapsto \|\omega\|_{\nu-\mathrm{Hol},[\mathrm{a,b}]}$ are measurable and $(C^{0,p-\mathrm{var}}([a,b],\mathbb{R}), \|\cdot\|_{p-\mathrm{var},[\mathrm{a,b}]})$ is a seperable Banach space, the map

$$(C^{0,p-\mathrm{var}}(\mathbb{R},\mathbb{R}),\mathcal{G}) \to (C^{0,p-\mathrm{var}}([a,b],\mathbb{R})\|\cdot\|_{p-\mathrm{var},[\mathrm{a,b}]})$$

$$\omega \mapsto \omega|_{[a,b]}$$

is measurable. Moreover, the Young integral satisfies the shift property with respect to θ, i.e.

$$\int_a^b x(u)d\omega(u) = \int_{a-r}^{b-r} x(u+r)d\theta_r\omega(u). \tag{2.5}$$

Theorem

*Under the assumptions in Theorem 2, there exists a unique solution to (**??**) with the initial value $x_0 \in \mathbb{R}^d$. Moreover, the solution $X : [t_0, t_0+T] \times [t_0, t_0+T] \times \mathbb{R}^d \times \Omega \to \mathbb{R}^d$ satisfies*

(i) $X(\cdot, a, x_0, \omega) \in C^{0,q-\mathrm{var}}([t_0, t_0+T], \mathbb{R}^d)$ with all ω.

(ii) $X(t,\cdot,\cdot,\cdot)$ is measurable w.r.t (a, x_0, ω).

(ii) $X(t, a, x_0, \cdot)$ is measurable w.r.t ω.

(iii) $X(t, a, x_0, \cdot)$ has finite expectation of order arbitrary $k \geq 1$ for all $t \in [t_0, t_0+T]$.

Definition (Cauchy operator)

Suppose that the assumptions of Theorem 2 are satisfied. For any $t_0 \leq t_1 \leq t_2 \leq t_0 + T$ the *random Cauchy operator* $\Phi_{t_1,t_2,\omega}$ of the SDE is defined as follows:

$$\Phi_\omega(t_1, t_2) : \mathbb{R}^d \to \mathbb{R}^d, \quad \omega \in \Omega,$$

is the mapping along trajectories from time moment t_1 to time moment t_2, i.e., for any vector $X_{t_1} \in \mathbb{R}^d$ we define $\Phi_\omega(t_1, t_2)X_{t_1}$ to be the vector $X_{t_2} \in \mathbb{R}^d$.

Definition

A family of continuous $\mathbb{R}^d-$value random mappings on $(\Omega, \mathcal{F}, \mathbb{P})$, $\Phi_\omega(s,t) : \mathbb{R}^d \to \mathbb{R}^d$ depending on $s, t \in [a, b] \subset \mathbb{R}$ is call a stochastic two-parameter flow of homeomorphisms of $\mathbb{R}^d$ on $[a, b]$ if $\Omega' \subset \Omega$ of full $\mathbb{P}$-measure:

(i) $\Phi_\omega(s,t) = \Phi_\omega(u,t) \circ \Phi_\omega(s,u)$ holds for all $s, t, u \in \mathbb{R}^+$;
(ii) $\Phi_\omega(s,s)$ is the identity map for all $s \in \mathbb{R}^+$;
(iii) the map $\Phi_\omega(s,t) : \mathbb{R}^d \to \mathbb{R}^d$ is an onto homeomorphism for all $s, t \in \mathbb{R}^+$;

Theorem

Suppose that the assumptions of Theorem 2 are satisfied. Then, the equation (2.1) *generates a stochastic two-parameter flow of linear operators of* $\mathbb{R}^d$ *by means of its Cauchy operators.*

Theorem (Generation of stochastic two parameter flow)

Suppose that the assumptions of Theorem 2 are satisfied. The family of random Cauchy operators of the SDE generates a stochastic two-parameter flow of homeomorphisms of $\mathbb{R}^d$ *on* $[0, T]$*. Namely, for* $0 \le t_1 \le t_2 \le T$ *we define* $\Phi_\omega(t_1, t_2)$ *according to Definition 5 and setting* $\Phi_\omega(t_2, t_1) := \Phi_\omega(t_1, t_2)^{-1}$*, then the family* $\Phi_\omega(t_1, t_2), t_1, t_2 \in [0, T]$*, is a stochastic two-parameter flow of homeomorphisms of* $\mathbb{R}^d$ *on* $[0, T]$*.*

We assume for simplicity that

$$\|A\|_{\infty,[0,\infty)} < \infty, \quad \|C\|_{q-\mathrm{var},\delta,\mathbb{R}^+} := \sup_{0\le t-s\le\delta} \|C\|_{q-\mathrm{var},[s,t]} < \infty, \forall\delta > 0. \tag{3.1}$$

Recall that for a real function $h : \mathbb{R}^+ \to \mathbb{R}^d$ the *Lyapunov exponent of* h is the number (which could be ∞ or $-\infty$)

$$\chi(h(t)) := \limsup_{t\to\infty} \frac{1}{t} \log |h(t)|.$$

Lyapunov exponents and Lyapunov spectrum

Definition

(i) The extended-real numbers (real numbers or symbol ∞ or $-\infty$)

$$\lambda_k(\omega) := \inf_{V \in \mathcal{G}_{d-k+1}} \sup_{y \in V} \limsup_{t \to \infty} \frac{1}{t} \log \|\Phi_\omega(t_0, t)\|, \quad k = 1, \ldots, d, \tag{3.2}$$

are called Lyapunov exponents of the flow $\Phi_\omega(s,t)$. The collection $\{\lambda_1(\omega), \ldots, \lambda_d(\omega)\}$ is called Lyapunov spectrum of the flow $\Phi_\omega(s,t)$.
(ii) For any $u \in [t_0, \infty)$ the linear subspaces of $\mathbb{R}^d$

$$E_k^u(\omega) := \left\{ y \in \mathbb{R}^d \;\Big|\; \limsup_{t \to \infty} \frac{1}{t} \log \|\Phi_\omega(u,t)y\| \le \lambda_k(\omega) \right\}, \quad k = 1, \ldots, d, \quad \omega \in \Omega, \tag{3.3}$$

are called Lyapunov subspaces at time u of the flow $\Phi_\omega(s,t)$. The flag of nonincreasing linear subspaces of $\mathbb{R}^d$ $\mathbb{R}^d = E_1^u(\omega) \supset E_2^u(\omega) \supset \cdots \supset E_d^u(\omega) \supset \{0\}$ is called Lyapunov flag at time u of the flow $\Phi_\omega(s,t)$.

Theorem

i, Let $\Phi_\omega(s,t)$ be a two-parameter stochastic flow of linear operators of $\mathbb{R}^d$ on the time interval $[t_0,\infty)$. Then the Lyapunov exponents $\lambda_k(\omega)$, $k=1,\ldots,d$, of $\Phi_\omega(s,t)$ are measurable functions of $\omega\in\Omega$.
ii, For any $u\in[t_0,\infty)$, the Lyapunov subspaces $E_k^u(\omega)$, $k=1,\ldots,d$, of $\Phi_\omega(s,t)$ are measurable with respect to $\omega\in\Omega$, and invariant with respect to the flow in the following sense

$$\Phi_\omega(s,t)E_k^s(\omega)=E_k^t(\omega),\ \forall s,t\in[t_0,\infty),\omega\in\Omega,k=1,\ldots,d.$$

Definition

The Lyapunov spectrum, Lyapunov exponents and Lyapunov subspaces of the linear SDE driven by fBm are those of the stochastic two-parameter flow $\Phi_\omega(s,t)$.

Theorem

Let $\Phi_\omega(s,t)$ be the two parameter flow generated by the SDE driven by fBm $\{\lambda_1(\omega),\ldots,\lambda_d(\omega)\}$ be the Lyapunov spectrum of the flow $\Phi_\omega(s,t)$. Then the Lyapunov exponents $\lambda_k(\omega), k=1,\ldots,d$, can be computed via a discrete-time interpolation of the flow, i.e.

$$\lambda_k(\omega) := \inf_{V\in\mathcal{G}_{d-k+1}} \sup_{y\in V} \limsup_{\mathbb{N}\ni t\to\infty} \frac{1}{t}\log\|\Phi_\omega(t_0,t)y\|,\ k=1,\ldots,d. \tag{3.4}$$

Moreover, the random variables $\lambda_1(\omega),\ldots,\lambda_d(\omega)$ have finite moments of order p for any $p\geq 1$. Hence, in particular, they are finite almost surely.

Proposition

The Lyapunov exponents $\lambda_k(\omega)$ do not depend on t_0.

Lemma

Under the assumptions (3.1), Φ_ω satisfies the integrability condition, i.e. for any $t_0 > 0$ and fixed $T > 0$, and any $m \geq 1$

$$E\Big(\sup_{t\in[t_0,t_0+T]} \log^+ \|\Phi_\omega(t_0,t)^{\pm 1}\|\Big)^m \leq (2\eta)^m \Big[2^m + \Big(\frac{4M^*}{\mu}\Big)^{mp}\Big(T^{mp} + C_{\nu,H,m,p} T^{Hmp}\Big)\Big] \quad (3.5)$$

Lemma

Let $\Phi_\omega(s,t)$ be the stochastic two-parameter flow of linear operators of $\mathbb{R}^d$ on the time interval $[t_0,\infty)$ generated by the linear SDE driven by fBm. Then for any $T \in \mathbb{R}^+$ there exists a constant $M = M(T)$ such that for any $a \in [t_0,\infty)$ the following inequality holds

$$\mathbb{P}\Big(\{\omega \in \Omega \mid \sup_{a\leq t\leq a+T} \log \|\Phi_\omega(a,t)\| \geq c^{1/3}\}\Big) \leq \frac{M}{c^2}, \qquad \forall c \in \mathbb{R}^+. \quad (3.6)$$

Old results

- Autonomous SDE: the spectrum is measurable and invariant w.r.t. ergodic DS θ, thus nonrandom
- Nonautonomous SDE with standard Brownian noises: the spectrum is measurable w.r.t the tail event from the filtration of mutually independent σ algebra $\sigma(B_t, t \in [n, n+1])$, thus nonrandom by Kolmogorov zero-one law.
- Nonautonomous SDE with fractional Brownian noises: $\sigma(B_t^H, t \in [n, n+1])$ are not mutually independent!!!

One dimensional systems

consider the homogeneous one dimensional linear equation

$$dx(t) = a(t)x(t)dt + c(t)x(t)d\omega(t), \tag{4.1}$$

in which $\|a\|_{\infty,\mathbb{R}^+}, \|c\|_{q-\mathrm{var},\delta,\mathbb{R}^+} < \infty$ to assure the existence of the solution on $\mathbb{R}^+$.

Proposition

The Lyapunov spectrum of 4.1 is nonrandom.

Triangular systems

we consider the system

$$dX(t) = A(t)X(t)dt + C(t)X(t)dB_t^H(\omega) \tag{4.2}$$

in which, $X = (x_1, x_2, ..., x_d)$, $A = (a_{ij}(t))$, $C = (c_{ij}(t))$ are d dimensional upper triangular matrix of coefficient functions satisfy the condition (3.1). We will show by inductive that the Lyapunov spectrum of system 4.2 is $\{\overline{a}_{kk}, 1 \leq k \leq d\}$ with $\overline{a}_{kk} = \lim_{t\to\infty} \frac{\int_0^t a_{kk}(s)ds}{t}$ provided that the limit is well-defined and exact.

Theorem

If there exists $\overline{a}_{kk} := \lim \frac{1}{t}\int_0^t a_{kk}(s)ds < \infty,\ k = \overline{1,d}$ *then the spectrum of system* (4.2) *is nonrandom and given by*

$$\{\overline{a}_{11}, \overline{a}_{22}, \ldots, \overline{a}_{dd}\}.$$

Lyapunov regularity

Definition

Let $\Phi_{s,t}(\omega)$ be a two-parameter stochastic flow of linear operators of $\mathbb{R}^d$ and $\{\lambda_1(\omega), \ldots, \lambda_d(\omega)\}$ be the Lyapunov spectrum of $\Phi_{s,t}(\omega)$. Then the non-negative $\bar{\mathbb{R}}$-valued random variable

$$\sigma(\omega) := \sum_{k=1}^{d} \lambda_k(\omega) - \underline{\delta}(\omega) = \sum_{k=1}^{d} \lambda_k - \liminf_{t\to\infty} \frac{1}{t} \log |\det \Phi_{0,t}(\omega)|$$

is called coefficient of nonregularity of the two-parameter flow $\Phi_{s,t}(\omega)$.

The coefficient of nonregularity of the linear SDE is, by definition, the coefficient of nonregularity of the two-parameter stochastic flow.

A two-parameter flow is called Lyapunov regular if its coefficient of nonregularity equals 0 identically. A linear SDE is called Lyapunov regular if its coefficient of nonregularity equals 0.

Proposition

(i) If a two-parameter linear stochastic flow $\Phi_{s,t}(\omega)$ *is Lyapunov regular then its determinant* $\det \Phi_{0,t}(\omega)$ *has exact Lyapunov exponent for all* $\omega \in \Omega$*.*

(ii) If a two-parameter flow $\Phi_{s,t}(\omega)$ *is Lyapunov regular then any trajectory has exact Lyapunov exponent.*

(iii) If the linear SDE is Lyapunov regular then any solution has exact Lyapunov exponent.

Lemma

(Stochastic Perron Theorem.) *Let* $\alpha_1 \geq \cdots \geq \alpha_d$ *and* $\beta_1 \leq \cdots \leq \beta_d$ *be the Lyapunov spectrum of the system and its adjoint respectively. Then the system is Lyapunov regular if and only if* $\alpha_i + \beta_i = 0$ *for all* $i = 1, \ldots, d$*.*

Theorem

Suppose that the matrices $A(t), C(t)$ are upper triangular and satisfy assumption (3.1)*. Then the system* (4.2) *is Lyapunov regular if and only if there exists* $\lim \frac{1}{t}\int_{t_0}^{t} a_{kk}(s)ds,\ k = \overline{1,d}$

Open problems

- Lyapunov regular systems possess nonrandom Lyapunov spectrum ?
- Counterexample ?

Thank you for your attention!